American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Junior Division

1. Boolean Algebra

Which ordered triple makes the following Boolean expression FALSE?

$$A(\overline{B + C}) + \overline{ABC}$$

A. (1,0,1)
B. (0,0,0)
C. (1,1,0)
D. (1,1,1)
E. (0,0,1)

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{AB}\,(\overline{A + C}) + \overline{C}$$

A. $\overline{A}$
B. $\overline{B}$
C. $\overline{C}$
D. $\overline{A} * \overline{C}$
E. $\overline{A} * \overline{B} * \overline{C}$

3. Data Structures

Build a binary search tree for the string (Left subtree $\leq$ root $<$ right subtree)

MASSACHUSETTS

How many nodes have exactly one child?

A. 3
B. 4
C. 5
D. 6
E. 9

4. Data Structures

Given an initially empty stack and the following commands on the stack, what will be popped next?

```
PUSH("B")
PUSH("O")
x = POP()
PUSH("S")
x = POP()
PUSH("T")
PUSH("O")
x = POP()
x = POP()
PUSH("N")
x = POP()
```

A. S
B. T
C. O
D. N
E. B

5. What Does This Program Do? (Arrays)

What is the output when the following program is run using the following values of array *arr*? Note that arr(0) has a value of 1.

1	4	9	16	25

```
num = 30
count = -1
while count != 3
     count = 0
     for y = 0 to 4
          if num % arr(y) == 0 then
               count = count + 1
          end if
     next y
     if count != 3 then
         num = num + 1
     end if
end while
output num
```

A. 32
B. 24
C. 36
D. 38
E. 40

6. What Does This Program Do? (Arrays)

What is the output when the following program is run using the following values of array *arr*? Note that arr(0,0) has a value of 3.

3	2	6	4	12	5
1	7	8	3	5	10
6	3	14	13	7	4

```
p = 1
for x = 0 to 5
     m = 2
     for y = 0 to 1
          if arr(y,x) < arr(m,x) then
               m = y
          end if
     next y
     if arr(m,x) < 5 then
          p = p * arr(m,x)
     end if
next x
output p
```

A. 10
B. 12
C. 18
D. 24
E. 30

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1. 布尔代数

请问哪一个有序三元组可以使得下列表达式的结果为**假**？

$$A(\overline{B\ +\ C})\ +\ \overline{ABC}$$

A. (1,0,1)
B. (0,0,0)
C. (1,1,0)
D. (1,1,1)
E. (0,0,1)

2. 布尔代数

请简化**下列布尔表达式**

$$\overline{AB}\ (\overline{A\ +\ C})\ +\ \overline{C}$$

A. $\overline{A}$
B. $\overline{B}$
C. $\overline{C}$
D. $\overline{A} * \overline{C}$
E. $\overline{A} * \overline{B} * \overline{C}$

3. 数据结构

为**以下**字符串构建一棵二叉搜索树。（左子树 ≤ 根 、右子树 > 根）

MASSACHUSETTS

请问该树中有多少个结点恰好只有一个子结点？

A. 3
B. 4
C. 5
D. 6
E. 9

4. 数据结构

给定一个初始为空的栈，以及对该栈执行的下列操作指令后，**请问**接下来将会弹出哪个元素？

```
PUSH("B")
PUSH("O")
x = POP()
PUSH("S")
x = POP()
PUSH("T")
PUSH("O")
x = POP()
x = POP()
PUSH("N")
x = POP()
```

A. S
B. T
C. O
D. N
E. B

5. 程序理解（数组）

当使用数组 *arr* 的下列数值运行**以下**程序时，**请问**输出结果是什么？
注意：arr(0) 的值为 1。

1	4	9	16	25

```
num = 30
count = -1
while count != 3
     count = 0
     for y = 0 to 4
          if num % arr(y) == 0 then
               count = count + 1
          end if
     next y
     if count != 3 then
         num = num + 1
     end if
end while
output num
```

A. 32
B. 24
C. 36
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E. 40

6. 程序理解（数组）

当使用数组 *arr* 的下列数值运行**以下**程序时，**请问**输出结果是什么？
注意：arr (0,0) 的值为 3。

3	2	6	4	12	5
1	7	8	3	5	10
6	3	14	13	7	4

```
p = 1
for x = 0 to 5
     m = 2
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     next y
     if arr(m,x) < 5 then
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```

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